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# Sales Assist

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## Documentation

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Team 3  
Member A  
Member B  
Member C  
Member D  
Member E

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# Index

|                                     |    |
|-------------------------------------|----|
| Change log.....                     | 3  |
| Requirements Analysis.....          | 4  |
| User Story.....                     | 5  |
| Key Technical Challenges .....      | 6  |
| Key Risks .....                     | 6  |
| Competition.....                    | 6  |
| What's New? .....                   | 7  |
| Workflow.....                       | 8  |
| Use Cases .....                     | 9  |
| Sales assist.....                   | 9  |
| Server .....                        | 11 |
| Use Case Diagrams .....             | 12 |
| System Sequence Diagram.....        | 13 |
| Domain Model.....                   | 17 |
| Test Cases .....                    | 18 |
| Test Cases for Iteration 1 .....    | 18 |
| Test Cases for Iteration 2.....     | 20 |
| Test Cases for Iteration 3.....     | 21 |
| Test Cases for Final Iteration..... | 23 |

# Change log:

## Iteration 1:

- Created Documentation files in multiple documents
- Created Documents for User story, Requirements, Use Cases, etc.

## Iteration 2:

- Updated Test cases
- Updated Requirements.

## Iteration 3:

- Integrated Documents into one file.
- Updated Test Cases.

## Final Iteration:

- Improved geo-location and maps
- Updated Test Cases
- Updated user cases, workflow, and sequence diagram.

# Requirements Analysis:

- ▶ **Customer:** As a salesperson, he wants to present to his customers how much benefit they could get if they pay with him. In order to impress the customers, the salesperson wants to show the result with histogram and pie-chart.

On the other hand, as a manager, he wants to track the salesperson's location, control the access to the application by the salesperson, and receive the customer's information from the sales.

- ▶ **Functional:** According to the customer's requirements, the application should at least include the following functions:
  - ▶ Functions to allow the salesperson or manager to log in the application.
  - ▶ Functions to allow the salesperson to input profiles of different customers.
  - ▶ Functions to enable the salesperson to input the business data from his customers, such as the type of credit card, the total amount of bills, the number of transactions, and percent of fees.
  - ▶ Functions to compute the benefit.
  - ▶ Functions to display the pie chart and histogram and synchronously renew the diagram when the business data is changed.
  - ▶ Functions to display the salesperson's location on the map.
  - ▶ Functions to allow manager to view the customer's information.
- ▶ **Environment:**
  - ▶ Application is to be interactive web application.
  - ▶ Must run on Android Tablet.
  - ▶ Must run with GPS service.

# User Story:

‘User Story is one or more sentences in the everyday or business language of the end user that captures what the user wants to achieve’ - **Wikipedia**

- As a Salesperson, I want to enter the customer details
- As a Salesperson, I want to enter the Business Values of the customer.
- As a Salesperson, I want to impress the customer by the look and feel of the application.
- As a Salesperson, I want to generate charts to help explain the benefits of our scheme to the customer
  - As a Salesperson, I want to generate a Histogram comparing the Transaction fees paid by the customer currently against the Transactions fees the customer will be required to pay with us.
  - As a Salesperson, I want to generate a Pie-chart comparing the different Monthly volumes for each of the credit card services.
- As a Manager, I want to track the Salesperson’s location and the Time spent with the customer.
- As a Manager, I want to control access to the application by the salesperson.
- As a Manager, I want to receive the customer’s information from the sales.

# Key Technical Challenges

- ▶ Process Real Time Data
  - ▶ Use Javascript libraries such as jQuery to handle events and process the Data.
- ▶ Charts
  - ▶ Multiple API's available to generate charts from given data
  - ▶ Eg: jqPlot, Sencha charts, Google charts etc.,
- ▶ Application-Server Communication
  - ▶ Server Application to be programmed using PHP.
- ▶ Divert to Setting
  - ▶ Can be done by calling the Location activity. But it will make the code specific to android. PhoneGap does not provide API.

# Key Risks

- ▶ Offline Data
- ▶ User Authentication
- ▶ JavaScript API's used may not be compatible with Android or other mobile devices.

# Competition

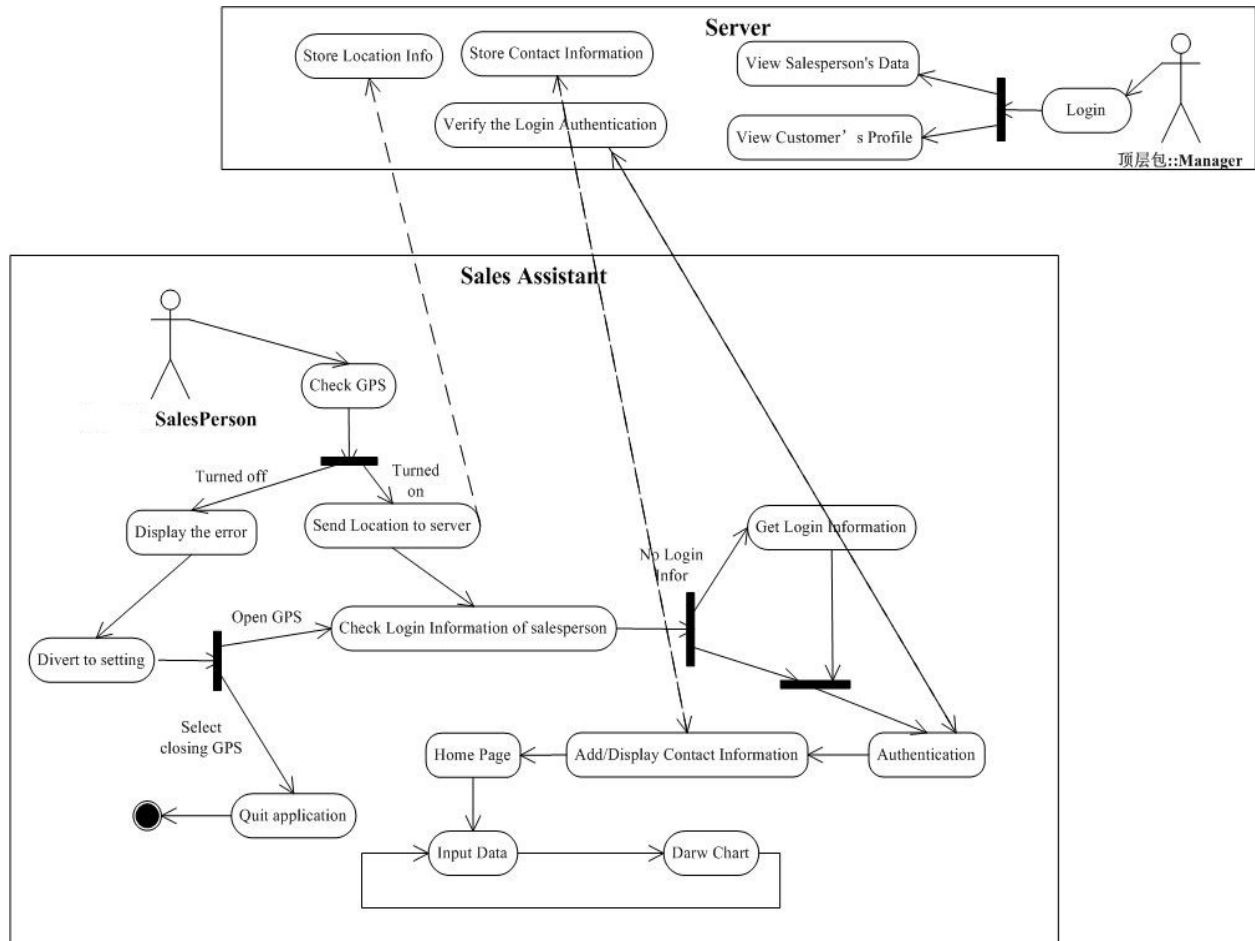
- ▶ Roam bi → business app for iPhone. Generates Business Analytics, Charts. Not a Sales App.
- ▶ Mobile bi → iPad app to draw charts, Not a Sales App.
- ▶ Merchant Genius → Visually Striking, Sales Application for Mac
- ▶ Proposable.com → Allows custom user configurations, Online Sales App, Not Free
- ▶ Mimiran Sales Compass → Complex, PC Application, Not Free

## What's New?

- ▶ Many Applications exist for PC, Mac, iOS.
- ▶ No Applications exist for Android
- ▶ Location data of the salesperson will be returned to the server
- ▶ Displays Charts to visually depict benefits for the customer

Custom Built to aid the Sales force of Credit Card Processing Companies.

# Workflow:



# Use Cases:

## Sales assist

**Actor:** Salesperson.

Use cases: Send Data, Edit Customer's Profile, and Generate Charts.

Actor Description: The salesperson is a user that interacts with the application (Sales Assist) to enter a business's information and generate pie charts and histograms.

Use Case Description:

1. Send Data:

- The Salesperson enters his log in information in the app.
- The System authenticates the user and sends the location data to the server.

Extended use case:

| Actor :Salesperson                                                               | System : Sales Assist                                                            |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1. TUCBW The user enters the log in information in the app                       | 2. *The System authenticates the user and sends the location data to the server. |
| 3. TUCEW The user gets a notification that the data has been sent to the server. |                                                                                  |

| Actor :Salesperson                                                               | System : Sales Assist                                                            |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1. TUCBW The user logs out of the application.                                   | 2. *The System sends the Customer data and the user location data to the server. |
| 3. TUCEW The user gets a notification that the data has been sent to the server. |                                                                                  |

## 2. Edit Customer's Profile:

- The user enters the business details like business name, point of contact, expenditure from various credit cards, etc. of the customer (A business owner).

Extended use case:

| Actor :Salesperson                                                  | System : Sales Assist                                   |
|---------------------------------------------------------------------|---------------------------------------------------------|
| 1. TUCBW The user logs in to the application.                       | 2. The System displays the Homepage.                    |
| 3. The user enters and edits the customer information.              | 4. *The System saves the data to be sent to the server. |
| 5. TUCEW The user gets a notification that the data has been saved. |                                                         |

## 3. Generate Charts:

- The application uses part of the entered information (Details of incomes via various credit card, processing fee charged by individual banks, etc.) to generate histograms and pie charts.

Extended use case:

| Actor :Salesperson                                             | System : Sales Assist                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1. TUCBW The user enters the business details of the customer. | 2. *The System uses the data for calculating and generating a pie chart and a histogram. |
| 3. TUCEW The user views the displayed charts.                  |                                                                                          |

# Server

**Actor:** Manager.

Use cases: Edit Salesperson's profile, View Salesperson's Location, View Customer's data.

Actor Description: The Manager is a user that interacts with the application (Server) to retrieve information entered by the salesperson, view location information for monitoring the salesperson's activity, authorize users (Salespersons) for using application, etc.

Use case Description:

1. Edit Salesperson's profile:

- The manager edits and modifies the salesperson's profile (including authorization details).

Extended use case:

| Actor : Manager                                                                       | System : Server                                              |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1. TUCBW The manager logs-in to the Server and clicks the "Manage Salesperson" link.. | 2. The server displays the "Manage Salesperson" page.        |
| 3. The manager selects and edits the profile of one of the Salesperson.               | 4. *The system saves the profile of the selected salesperson |
| 5. TUCEW The manager sees a notification that the profile has been changed.           |                                                              |

2. View Salesperson's Location:

- The manager logs-in to the Server and clicks the "Locate Salesperson" link.
- The System returns a Map with the locations of all the salespersons.
- The manager views the location of the salespersons.

Extended use case:

| Actor : Manager                                                                      | System : Server                                                          |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1. TUCBW The manager logs-in to the Server and clicks the "Locate Salesperson" link. | 2. *The System returns a Map with the locations of all the salespersons. |
| 3. TUCEW The manager views the location of the salespersons                          |                                                                          |

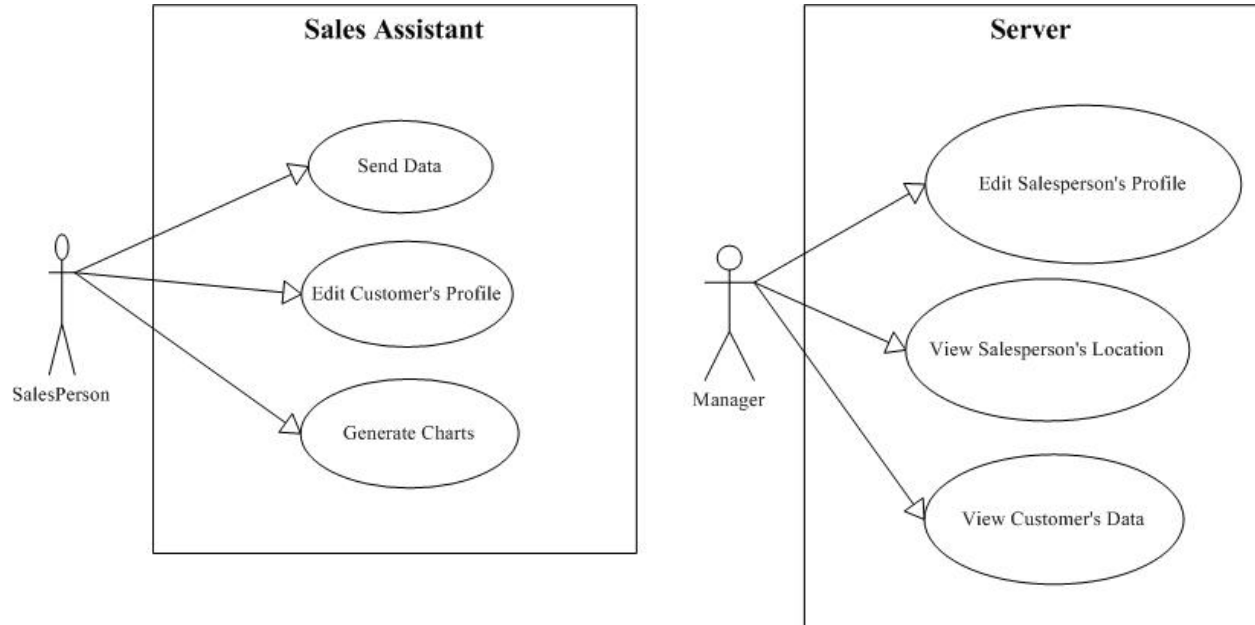
3. View customer's data:

- The manager logs-in to the Server and clicks the “View Customers” link.
- The server displays the “Customer details” page.
- The manager views the customer details.

Extended use case:

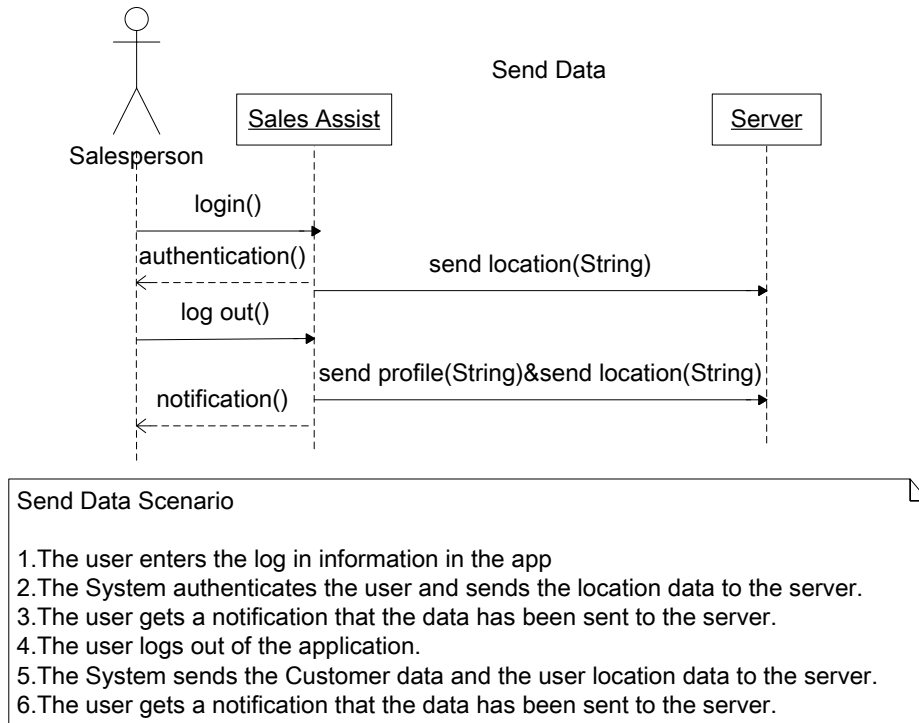
| Actor : Manager                                                                  | System : Server                                      |
|----------------------------------------------------------------------------------|------------------------------------------------------|
| 1. TUCBW The manager logs-in to the Server and clicks the “View Customers” link. | 2. *The server displays the “Customer details” page. |
| 3. TUCEW The manager views the customer details.                                 |                                                      |

## Use Case Diagrams:

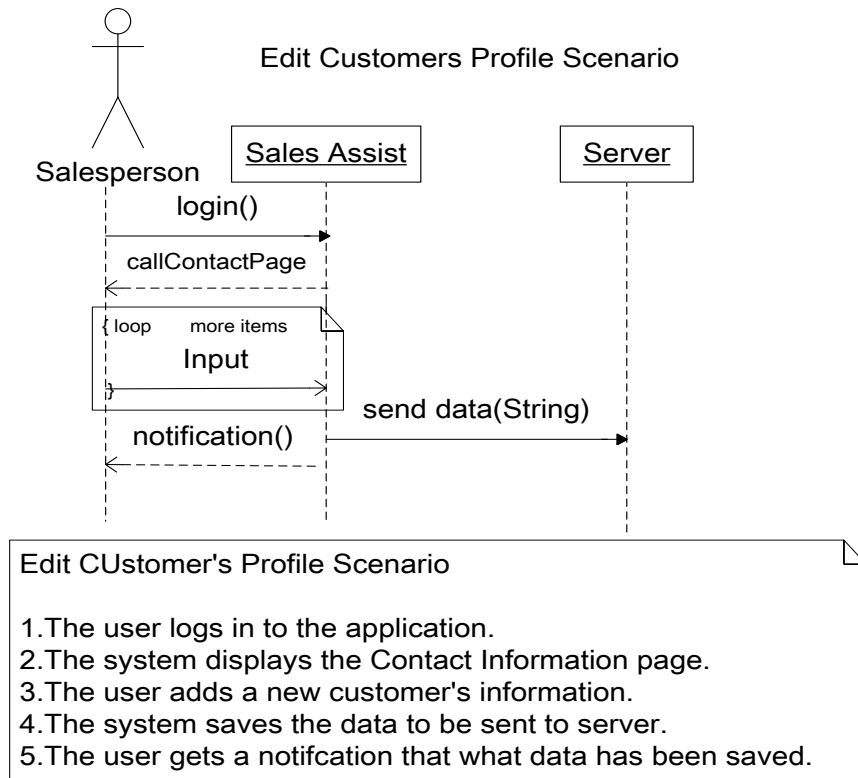


# System Sequence Diagram:

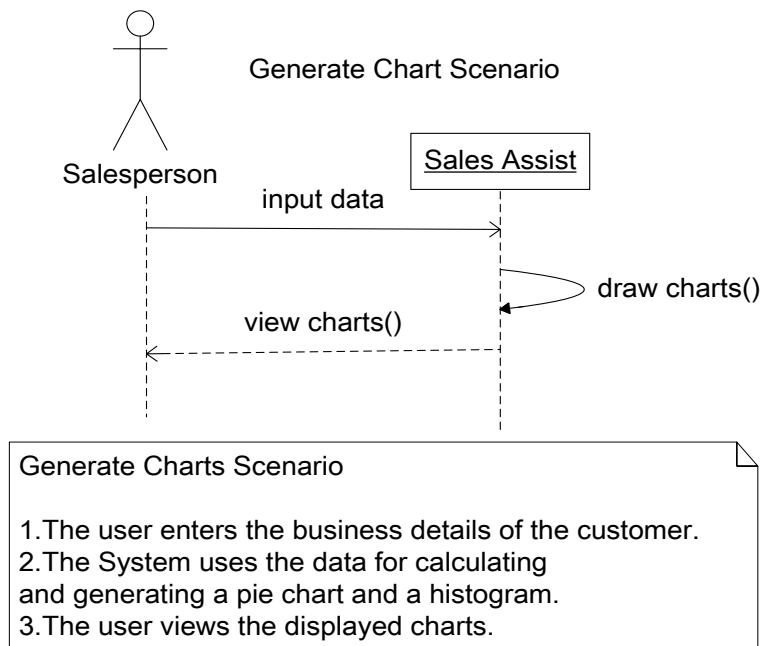
- Sales Assist
  - Send Data Scenario



- Edit Customer's Profile Scenario

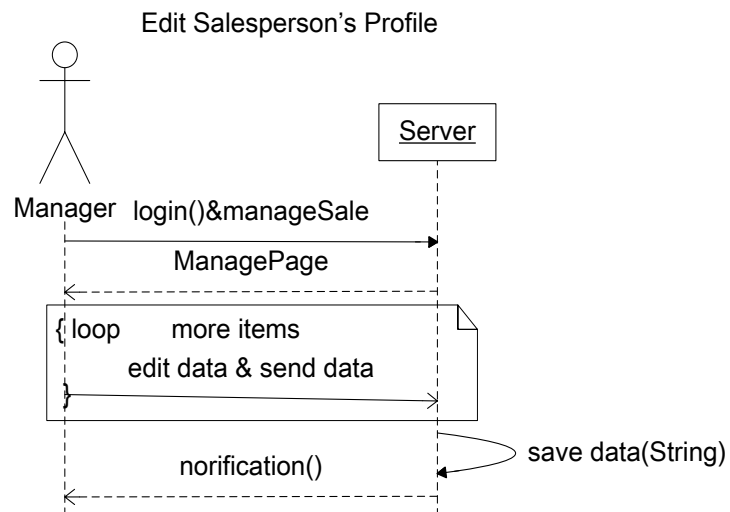


► **Generate Charts Scenario**



## ► Server

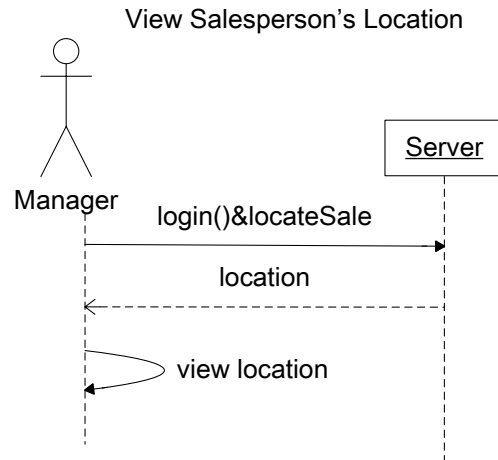
### ► Edit Salesperson's Profile Scenario



#### Edit Salesperson's Profile

1. The manager logs-in to the Server and clicks the "Manage Salesperson" link.
2. The server displays the "Manage Salesperson" page.
3. The manager selects and edits the profile of one of the Salesperson.
4. The manager sees a notification that the profile has been changed.

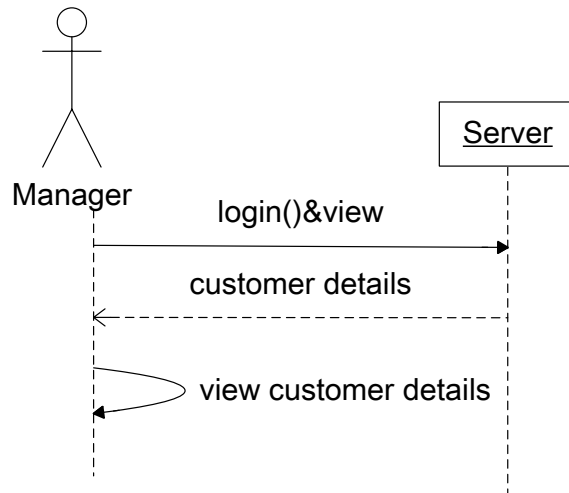
### ► View Salesperson's Location



**View Salesperson's Location**

- 1.The manager logs-in to the Server and clicks the "Locate Salesperson" link.
- 2.The server returns a Map with the locations of all the salespersons.
- 3.The manager views the location of the salespersons

► View Customer's Data

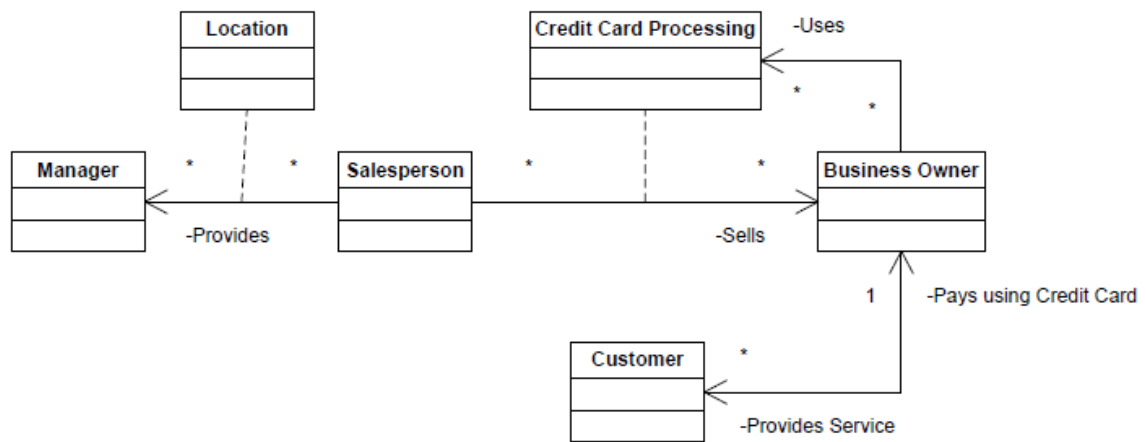


**View Customer's Data Secenario**

- 1.The manager logs-in to the Server and clicks the "View Customers" link.
- 2.The server displays the "Customer details" page.
- 3.The manager views the customer details

# Domain Model:

Sales Assist – Domain Model



# Test Cases:

## Test Cases for Iteration 1:

- **Target Users:** Salespersons.
- **What is covered in the tests:**
  1. The basic functionalities of the application.
  2. The usability of the application.
  3. The validation of the data provided by the users.
- **Test Cases:**
  1. **Test case id:** T1.  
**Test Case Description:** This test case checks if the Selection Box allows the selection of only one item from the drop down menu.  
**Action:** d\_select (Selection of the credit card name).  
**Related Requirements:** The user must be logged in and authenticated.  
**Expected Result:** The selection box must allow the selection of only one item.  
**Test Result:** The selection box allows the selection of only one item.  
**Status:** Pass.
  2. **Test case id:** T2.  
**Test Case Description:** This test case checks if the input field allows the use of only numeric characters as an input.  
**Action:** input (Used as the input field for: Amount per Month, Transactions per Month, and Fees in Percent).  
**Related Requirements:** The user must be logged in and authenticated.  
**Expected Result:** The input field must allow only numeric data as a valid input.  
**Test Result:** The input field allows only numeric data as a valid input.  
**Status:** Pass.
  3. **Test case id:** T3.  
**Test Case Description:** This test case checks if the pie chart gets updated dynamically as soon as the data in the input fields **Amount per Month** is manipulated.  
**Action:** drawPieChart.  
**Related Requirements:** The user must be logged in and authenticated.  
**Expected Result:** The pie chart should change dynamically as one of the amount fields is modified.  
**Test Result:** The pie chart changes dynamically as data in one of the input fields is modified.  
**Status:** Pass.
  4. **Test case id:** T4.

**Test Case Description:** This test case checks if the pie chart gets updated dynamically as soon as the data in the input fields **Amount per Month** and **Fees in Percent** is modified or updated.

**Action:** drawHistogram.

**Related Requirements:** The user must be logged in and authenticated.

**Expected Result:** The histogram must change dynamically as data in one of the input fields is modified.

**Test Result:** The histogram changes dynamically as data in one of the input fields is modified.

**Status:** Pass.

5. **Test case id:** T5.

**Test Case Description:** This test case checks if the Total Business (Amount) gets calculated dynamically as soon as the data in the input fields **Amount per Month** is modified or updated.

**Action:** Total Business.

**Related Requirements:**

- a. The user must be logged in and authenticated.
- b. There must be numeric data present in the **Amount per Month** column of at least one field.

**Expected Result:** The amount shown by Total Business must be the Total Sum of the values entered in the Amount fields.

**Test Result:** The amount shown by Total Business is the Total Sum of the values entered in the Amount fields

**Status:** Pass.

6. **Test case id:** T6.

**Test Case Description:** This test case checks if the Average Ticket Size (Amount) gets calculated dynamically as soon as the data in the input fields **Amount per Month** and **Transactions per Month** is modified or updated.

**Action:** Average Ticket Size.

**Related Requirements:**

- a. The user must be logged in and authenticated.
- b. There must be numeric data present in the **Amount per Month** column of at least one field.
- c. There must be numeric data present in the **Transactions per Month** column of the same field.

**Expected Result:** The amount shown by Average Ticket Size must be dynamically calculated from the data present in the input fields.

**Test Result:** The amount shown by Average Ticket Size is dynamically calculated from the data present in the input fields.

**Status:** Pass.

## Test Cases for Iteration 2:

- **Target User:** Salespersons.
- **What is covered in the tests:**
  1. The login of the application.
  2. The notification of location status of the application.

- **Test Cases:**

1. **Test case id:** T1.

**Test Case Description:** This test case checks if the user's login information could be validation.

**Action:** validate (Validate the user name and password)

**Expected Result:** Without inputting user name and password, the user could not log in the application. The application could notify user to input his login information.

**Test Result:** With validated information, user could log in the application.

**Status:** Pass.

2. **Test case id:** T2.

**Test Case Description:** This test case checks if the user's login information matches the record in the database.

**Action:** Login (Compare the input information with the record in the database)

**Related Requirements:** The user's login information has been created in the database.

**Expected Result:** If the input information matches the record in the database, the user then could view the main page. Otherwise, the application could alert invalidation to user.

**Test Result:** With correct login data, the user could view the main page. Without correct login data, the user could see an alert window.

**Status:** Pass.

3. **Test case id:** T3.

**Test Case Description:** This test case checks if the location of user is available.

**Action:** getLocation

**Related Requirements:** The cell phone or tablet should have GPS service.

**Expected Results:** If the GPS device turns off, the application alerts user to turn it on.

**Test Result:** If GPS service turns on, the user directly logs in the application. Otherwise, the application sends an alert to the user.

**Status:** Pass.

## Test Cases for Iteration 3:

- **Target Users:** Salespersons and Manager.
- **What is covered in the tests:**
  1. The basic functionalities of the new web page Contact Information.
  2. The function of saving data from application to the server.
  3. The basic functionality of the geo-location on the server side.
- **Test Cases:**
  1. **Test case id:** T1.  
**Test Case Description:** This test case checks if the input field allows the user of text, email, and numeric characters as an input.  
**Action:** input (Used as the input field for: Company Information and Contact Person Information).  
**Related Requirements:** The salesperson must be logged in and authenticated.  
**Expected Result:** The input field should allow to input the specific characters.  
**Test Result:** The input field allows specific inputting characters.  
**Status:** Pass.
  2. **Test case id:** T2.  
**Test Case Description:** This test case checks if the Next button allows the user to access to the main page.  
**Action:** Forward().  
**Related Requirements:** The user must be logged in and authenticated.  
**Expected Result:** The Next button must link to the main page.  
**Test Result:** The user could view the main page after click the Next button  
**Status:** Pass.
  3. **Test case id:** T3.  
**Test Case Description:** This test case checks if the Cancel button allows the user to clear the input field  
**Action:** Cancel().  
**Related Requirements:** The user must be logged in and authenticated.  
**Expected Result:** All input field should be cleared.  
**Test Result:** After click the Cancel button, the entire input field has been cleared.  
**Status:** Pass.
  4. **Test case id:** T4.  
**Test Case Description:** This test case checks if the Save button sends the contact information to the server and the server  
**Action:** Save().  
**Related Requirements:** The user must be logged in and authenticated.  
**Expected Result:** All the data are sent to the server, and the user knows what have been sent to the server.

**Test Result:** The application sends a notification to the user that data have been sent and display the sent data.

**Status:** Pass.

5. **Test case id:** T5.

**Test Case Description:** This test case checks if the manger could view the geo-location information on the map

**Action:**

**Expected Result:** The manager could see all the customers' locations displayed on the map.

**Test Result:** All the customers' locations are displayed on the map. When the manager clicks the pin point, the customer's information will be displayed.

**Status:** Pass.

## Test Cases for Final Iteration:

- **Target User:** Salespersons and Manager

- **What is covered in the tests:**

1. The customer's information is saved to the database.
2. The location history list is shown on the map.

- **Test Cases:**

1. **Test case id:** T1.

**Test Case Description:** This test case checks if the Save button sends the contact information to the server and the server

**Action:** Save().

**Related Requirements:** The user must be logged in and authenticated.

**Expected Result:** All the data are sent to the server, and the user has the information about the data that has been sent to the server.

**Test Result:** The application sends a notification to the user that data have been sent and display the sent data.

**Status:** Pass.

2. **Test case id:** T2.

**Test Case Description:** This test case checks if the manager could view the location history when he click the marker on the map.

**Action:**

**Related Requirements:** The user must be logged in and authenticated.

**Expected Result:** The location history will be displayed on the left when user clicks the marker on the map.

**Test Result:** The websites lists the location history on the left of the map when clicking the marker.

**Status:** Pass

3. **Test case id:** T3.

**Test Case Description:** This test case checks if the colors of the credit cards are synchronized with that on the pie chart

**Action:**

**Related Requirements:** The user must be logged in and authenticated.

**Expected Result:** The colors of the characters are the same with that on the pie chart.

**Test Result:** The colors of credit cards in the form and charts are the same.

**Status:** Pass